**EUROPEAN HIGHER EDUCATION INSTITUTE**

**Python - Final Exam Project**

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# Open Question

A shop sells a range of mobile devices, SIM cards and accessories as shown in the table:

**Table - 1**

| Category | Item code | Description | Price ($) |
|---|---|---|---|
| Phone | BPCM | Compact | 29.99 |
| Phone | BPSH | Clam shell | 49.99 |
| Phone | RPSS | Robo phone - 5inch 64GB memory | 199.99 |
| Phone | RPLL | Robo phone - 6inch 256GB memory | 499.99 |
| Phone | YPLS | Y-phone standard 6 inch 64GB memory | 549.99 |
| Phone | YPLL | Y-phone deluxe 6 inch 256GB memory | 649.99 |
| Tablet | RTMS | RoboTab - 8 inch screen 64GB memory | 149.99 |
| Tablet | RTML | RoboTab - 10 inch screen 128GB memory | 299.99 |
| Tablet | YTLM | Y-tab standard - 10 inch screen 128GB memory | 499.99 |
| Tablet | YTLL | Y-tab deluxe - 10 inch screen 256GB memory | 599.99 |
| SIM card | SMNO | Sim free (no SIM card) | 0.00 |
| SIM card | SMPG | Pay as you go (with SIM card) | 9.99 |
| Case | CSST | Standard | 0.00 |
| Case | CSLX | Luxury | 50.00 |
| Charger | CGCR | Car | 19.99 |
| Charger | CGHM | Home | 15.99 |

Write a program algorithm for this shop.

- Your program or programs must include appropriate prompts for the entry of data; data must be validated on entry.
- Error messages and other output need to be set out clearly and understandably.
- All arrays, variables, constants and other identifiers must have meaningful names.

You will need to complete these three tasks.

### Task 1 – Setting up the system.

Write a program to:

- use appropriate data structures to store the item code, description and price information for the mobile devices, SIM cards and accessories;
- allow the customer to choose a specific phone or tablet;

- allow phone customers to choose whether the phone will be SIM Free or Pay As You Go;
- allow the customer to choose a standard or luxury case;
- allow the customer to choose the chargers required (none, one or both may be purchased);
- calculate the total price of this transaction;
- output a list of the items purchased and the total price.

**Task 2 – Allow a customer to order multiple mobile devices. Extend Task 1 to:**

- offer the customer the opportunity to purchase an additional mobile device;
- if required, perform bulleted steps 2 to 7 of Task 1 for each additional mobile device and calculate a running total for the customer;
- once no further devices are required, output the total the customer will need to pay.

**Task 3 – Offering discounts.**

Extend the program to allow a discount of 10% off the price of every additional phone or tablet purchased. Output the new total the customer will need to pay and the amount of money saved.

## Answer with Python File

This is the Google Drive link of the completed Python Final Exam Project of Çiçek Akkaya:

- https://drive.google.com/file/d/1fVhfu8psva-EYzfMw2x9fLV7Ct7Qy-Ek/view?usp=sharing